

Use-Case Flow Specification

Problem Statement #12: Blood Bank Inventory & Emergency Donor Matcher

Use Case: UC-01 — Submit Emergency Blood Request

Primary Actor: Emergency Requester

Supporting Actors: Blood Bank Manager, Donor, SMS Gateway (external system)

Preconditions

- The Emergency Requester is authenticated on the system.
- The blood bank inventory database and SMS gateway are online and reachable.
- At least one donor profile with a valid blood group and location exists in the system.

Postconditions

- Success: The request is logged with a status of "Fulfilled" or "Donors Notified", and matched donor responses are recorded against it.
- Failure: The request is logged with a status of "Escalated" or "Cancelled", and the Blood Bank Manager is notified.

Main Success Scenario

1. Emergency Requester logs in and selects "Request Emergency Blood."
2. System displays a form for blood group, component type, quantity, and hospital location.
3. Requester submits the request.
4. System validates the input and cross-references it against real-time inventory.
5. If on-hand inventory is insufficient, the system checks ABO/Rh compatibility and identifies eligible donors within a 10 km radius.
6. System sends SMS alerts to all compatible, eligible donors via the SMS Gateway.
7. Donors receive the alert and respond Confirm or Decline via a reply link/SMS keyword.
8. System records each donor's response in real time and updates the request's donor-status list.
9. Once enough donors confirm to meet the requested quantity, the system notifies the Blood Bank Manager and the Emergency Requester with donor details and estimated arrival.
10. Use case ends successfully; request status is set to "Fulfilled."

Alternate Flow — 6a: No Compatible Donors Found Within 10 km

- a. System automatically expands the search radius to 25 km and repeats donor matching.
- b. If still no eligible donors are found, the system escalates the request to partner blood banks in the network and alerts the Blood Bank Manager.
- c. Emergency Requester is shown an updated status ("Escalated") along with the expected response time.
- d. If no partner blood bank can fulfil the request within a defined time window, the Blood Bank Manager manually closes or reassigns the request, and the use case ends in failure.

Alternate Flow — 7a: Donor Declines or Does Not Respond

- a. If a donor declines, their status is set to "Declined" and they are removed from the active notification list for this request.
- b. If a donor does not respond within 15 minutes, the system marks them "No Response" and continues waiting for other donors without blocking the overall flow.
- c. If the confirmed donor count still falls short after the response window, the flow rejoins step 6a (radius expansion / escalation).

Requirements Table

Lab 1: Requirements Engineering & UML Use-Case Modelling — Problem Statement #12: Blood Bank Inventory & Emergency Donor Matcher
 Rows shaded blue (FR-001, NFR-001) are the instructor-provided baseline requirements. Remaining rows were authored for this problem statement.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Pass: Compatible donors notified within 30 seconds of emergency flag. Fail: Incompatible blood group alerted.	Given requirement (baseline). Ensures the core emergency-matching function works correctly and quickly.
FR-002	Functional	The system shall verify ABO and Rh(D) compatibility between the requested blood type and each candidate donor's blood type before adding that donor to the notification list.	High	Pass: In test runs covering all 8 blood group combinations, zero incompatible donors appear in the notification list. Fail: Any incompatible donor is included.	Prevents life-threatening transfusion errors caused by incorrect blood-type matching.
FR-003	Functional	The system shall allow a Blood Bank Manager to add, remove, or adjust inventory counts by blood group and component type (whole blood, plasma, platelets).	High	Pass: Inventory count reflects the manager's change within 2 seconds and is visible on the dashboard. Fail: Count remains stale after 2 seconds or does not match the entered value.	Accurate, current stock levels are required before the system can correctly decide whether an emergency donor search is needed.
FR-004	Functional	The system shall allow verified medical centres to log high-urgency blood requests, capturing the required quantity, urgency level (Immediate/Code Red vs. Scheduled Surgery), and patient blood profile.	High	Pass: A verified medical centre can submit a request that captures quantity, urgency level, and blood profile, and the system rejects submission if the requester is unverified or any required field is missing. Fail: A request is accepted without an urgency level/blood profile, or from an unverified medical centre.	Ensures accurate triage so that Immediate/Code Red requests are prioritised for matching and fulfillment ahead of scheduled, non-emergency requests.
FR-005	Functional	The system shall automatically flag blood units nearing expiry (within 48 hours) on the Blood Bank Manager's dashboard.	Medium	Pass: 100% of units with less than 48 hours to expiry are shown flagged (e.g., in red) on the dashboard. Fail: Any qualifying unit is left unflagged.	Reduces wastage of perishable blood units and supports proactive stock rotation.
NFR-001	Nonfunctional (Performance & Security)	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Pass: Benchmarking/load tests confirm target latency and no double-allocation under simulated peak concurrent load. Fail: Any test run produces a double-allocated unit or exceeds target latency.	Given requirement (baseline). Prevents double-booking of a single physical blood unit to two different requests.
NFR-002	Nonfunctional (Security & Privacy)	The system shall encrypt all donor personal and medical data (blood group, contact number, location) both in transit and at rest.	High	Pass: Security audit confirms TLS 1.2+ is used for all data in transit and AES-256 for all data at rest, with no field found in plaintext. Fail: Any donor data field is found unencrypted in transit or storage.	Protects sensitive personal health information and supports compliance with data-protection regulations.

Blood Bank Inventory & Emergency Donor Matcher - Use-Case Diagram

